Engineer to Excel

TECHNICAL PRESENTATION

Operations In A Modern Computer

CSA0406 – Operating System · CO1_AT3

Guided By

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Introduction

Operating System Functions and System Calls

1. An Operating System (OS) is system software.
2. It acts as a bridge between users and computer hardware.
3. It manages system resources efficiently.
4. It controls CPU, memory, files, and I/O devices.
5. System calls allow programs to communicate with the OS.

Major Functions of an Operating System

01

Process Management

Manages running programs and CPU scheduling.

02

Memory Management

Allocates and manages RAM.

03

File Management

Creates, stores, and manages files.

04

I/O Management

Controls input and output devices.

05

Resource Management

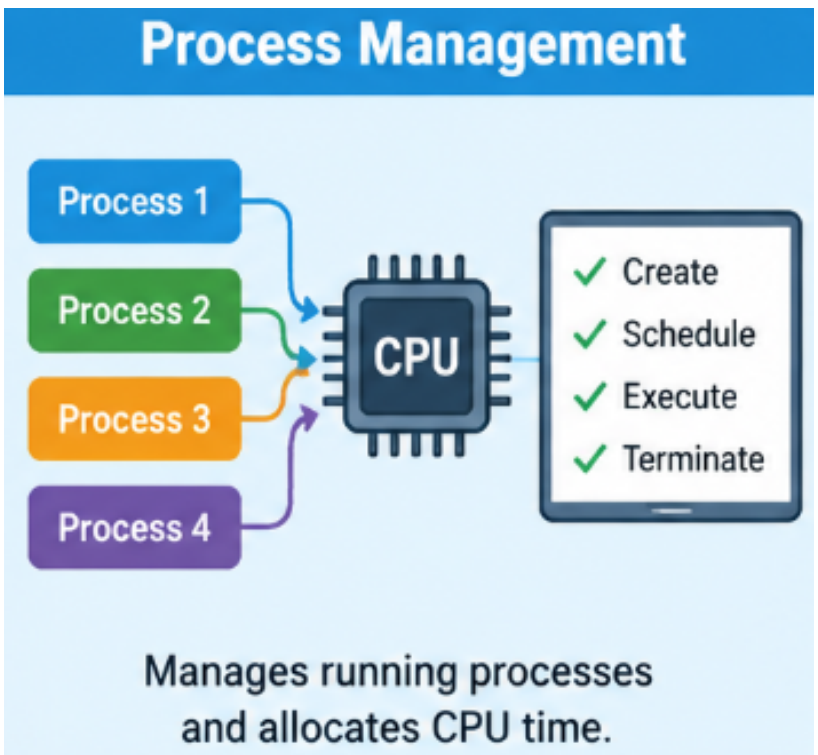
Allocates resources to different processes.

Importance

These functions improve performance, security, and efficient resource usage.

Process Management

- A process is a program that is currently running.
- OS creates, schedules, and terminates processes.
- CPU scheduling decides which process runs next.
- OS allows multiple processes to run efficiently.
- It also provides process synchronization.

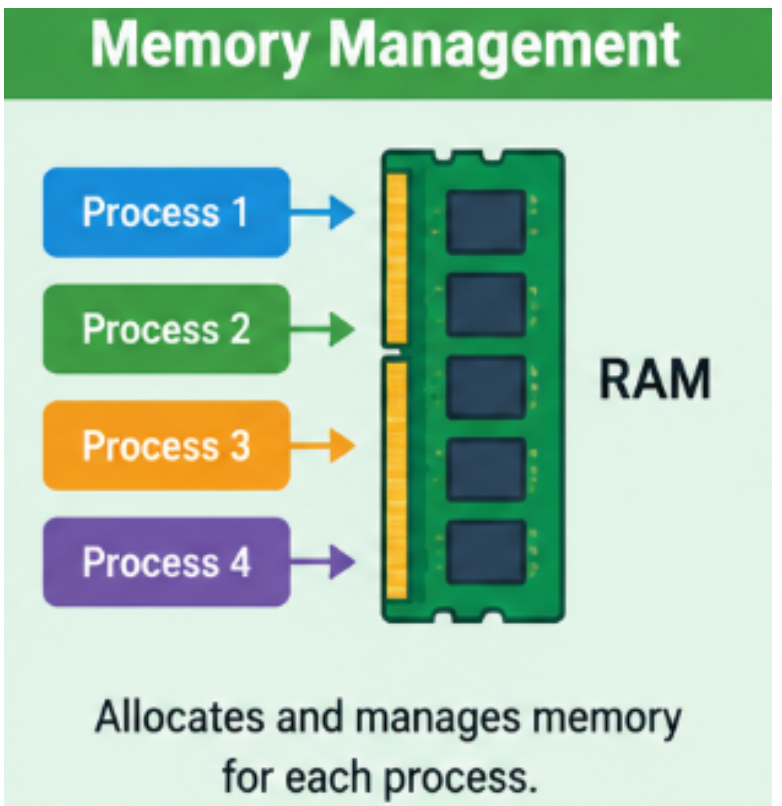


EXAMPLE SYSTEM CALLS

fork()	Creates a new process.
exec()	Runs a new program.
wait()	Waits for a process to finish.
exit()	Terminates a process.

Memory Management

- OS manages the computer's main memory (RAM).
- Allocates memory to running processes.
- Frees memory when it is no longer needed.
- Protects one process's memory from another.
- Supports virtual memory when required.

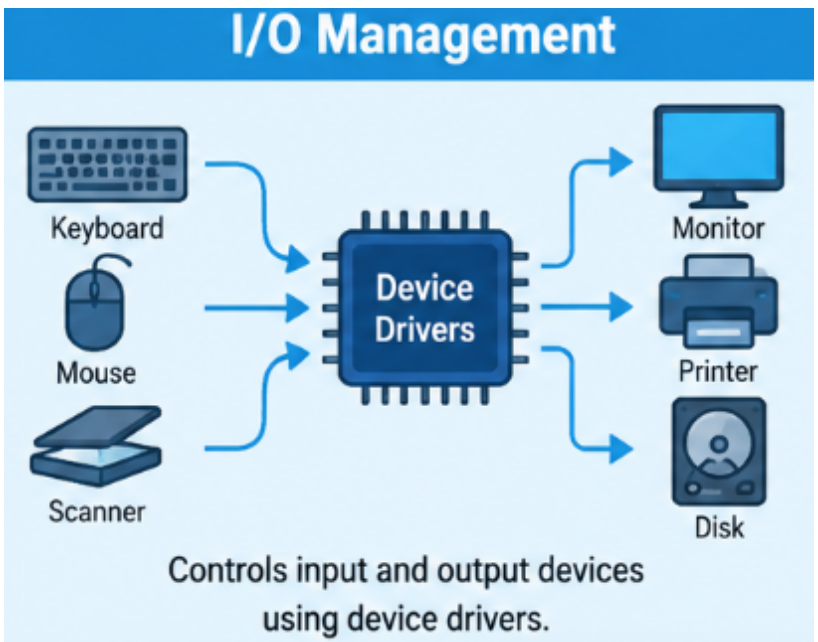


EXAMPLE SYSTEM CALLS

mmap()	Maps memory into a process.
munmap()	Releases mapped memory.
brk()	Changes the process memory size.
Goal	Efficient and safe use of memory.

I/O Management

- I/O means Input/Output.
- OS manages devices such as keyboard, mouse, disk, and printer.
- Device drivers help the OS communicate with hardware.
- OS controls reading and writing of data.
- It provides a common interface for different devices.



EXAMPLE SYSTEM CALLS

open()	Opens a file/device.
read()	Reads data.
write()	Writes data.
close()	Closes a file/device.

Advantages & Disadvantages

Advantages

Efficient Resource Management

Manages CPU, memory, and I/O effectively.

Multitasking

Allows multiple programs to run simultaneously.

Security

Protects data and system resources.

Disadvantages

High Resource Usage

OS consumes CPU and memory.

Security Vulnerabilities

Can be affected by malware and attacks.

System Failure

OS failure can affect the entire system.

Advanced OS Quiz – Part 1

Choose the correct option.

1. A running program needs CPU scheduling and later terminates itself. Which combination is correct?
A) File Management + open()
B) Process Management + exit()
C) Memory Management + munmap()
D) I/O Management + close()

2. A parent creates a child; the child runs a new program; the parent waits. Which sequence matches the presentation?
A) fork() → exec() → wait()
B) mmap() → brk() → munmap()
C) open() → read() → write()
D) wait() → fork() → exit()

3. Which OS function protects one process's memory from another while managing RAM and virtual memory?
A) I/O Management
B) File Management
C) Memory Management
D) Resource Management

4. A program creates a memory mapping and later releases that mapping. Which pair fits the presentation?
A) fork() + wait()
B) open() + close()
C) mmap() + munmap()
D) brk() + exit()

5. A process needs to change its memory size, not map or unmap a region. Which call matches the presentation?
A) brk()
B) mmap()
C) munmap()
D) wait()

Advanced OS Quiz – Part 2

Choose the correct option.

6. Which pair matches the presentation's description of common device access and hardware-specific communication?
A) Process Management + system calls
B) I/O Management + device drivers
C) Memory Management + virtual memory
D) File Management + CPU scheduling

7. A program accesses a file/device, reads data, writes data, then finishes using it. Which call order fits?
A) read() → open() → close() → write()
B) open() → read() → write() → close()
C) open() → fork() → wait() → close()
D) mmap() → read() → brk() → close()

8. Several processes are ready, and the OS must choose which process runs next. Which function handles this decision?
A) Memory Management
B) File Management
C) Process Management
D) I/O Management

9. The OS allocates CPU, memory, and I/O resources among processes. Which function most directly matches this responsibility?
A) Resource Management
B) File Management
C) Process Management
D) Device Drivers

10. An application needs OS services to communicate with the kernel rather than directly control hardware. What mechanism provides this communication?
A) Virtual memory
B) Device drivers
C) System calls
D) CPU scheduling

Conclusion

01 The OS is responsible for managing computer resources.

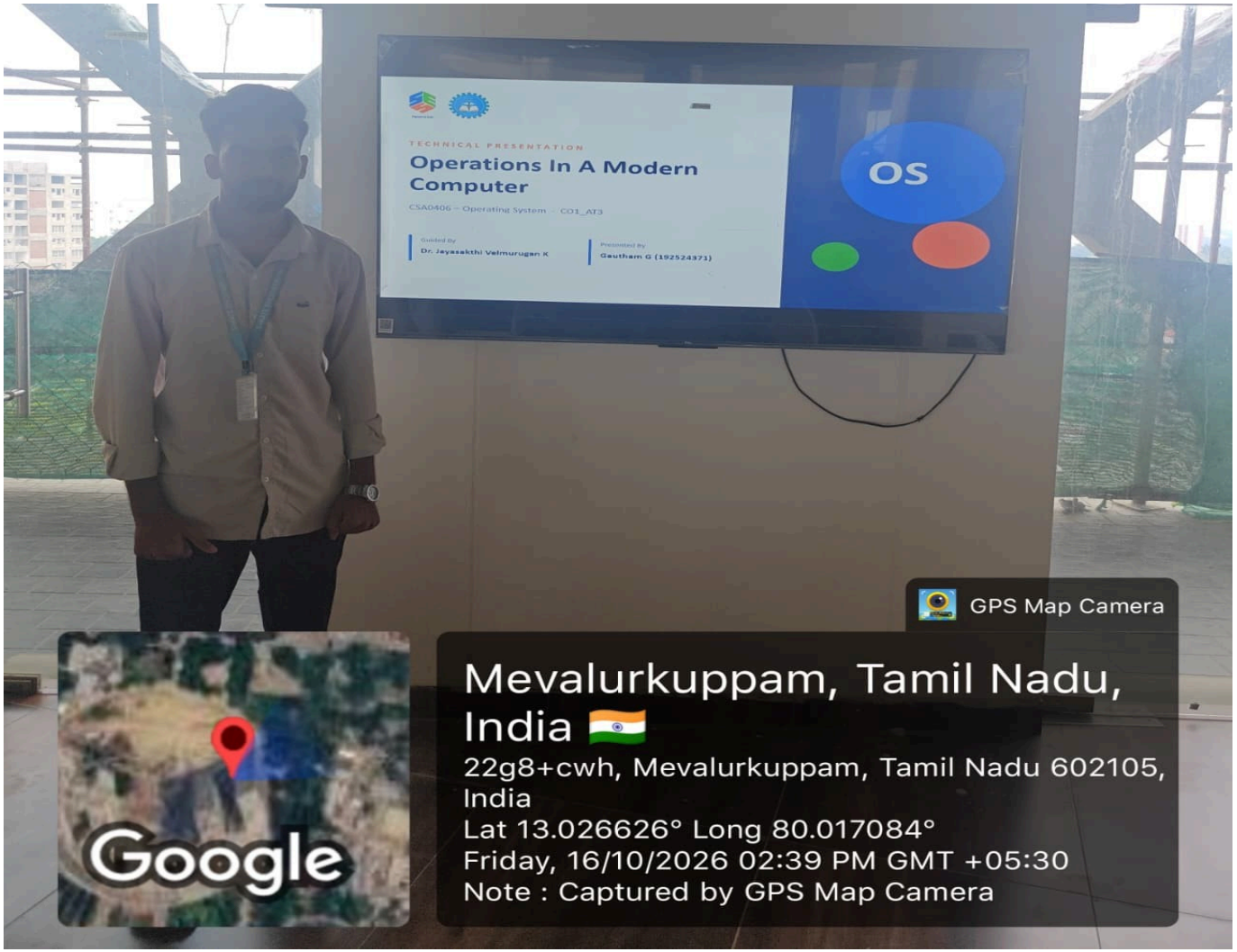
02 Process management controls running programs.

03 Memory management handles RAM efficiently.

04 I/O management controls input and output devices.

05 OS components work together to provide smooth system operation.

06 System calls provide communication between user programs and the kernel.


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Thank
you!